



Society of Petroleum Engineers

SPE-227178-MS

Breaking Down Complexity: Using Multiple Point Statistics to Improve Fracture Network Modeling in Carbonate Reservoirs

Neelesh Tripathi and Amr F. Elsharabasy, Saudi Aramco, Dhahran, Saudi Arabia; Kashif Saeed, Aspen Technology, Inc., Khobar, Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227178-MS](https://doi.org/10.2118/227178-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This study aims to advance carbonate reservoir modeling by employing Multiple Point Statistics (MPS), a geostatistical method designed to capture complex spatial correlations for simulating realistic fracture networks. Traditional Discrete Fracture Network (DFN) models often oversimplify fracture systems, leading to inaccuracies in predicting reservoir performance. By integrating high-fidelity digital outcrop data, seismic attributes, and the Direct Sampling (DS) method ([Mariethoz et al., 2010](#)), this research develops a robust MPS framework. Validation using the Teapot Dome dataset demonstrates that MPS reduces prediction errors in permeability, pressure profiles, and production rates by approximately 50% compared to DFN models. Reservoir simulations were conducted with the state-of-the-art reservoir simulator to further cement MPS's superiority, showing sustained pressure maintenance due to effective communication within hydrocarbon, hydrocarbon-water and within water-leg leading to 15–20% higher cumulative oil production due to enhanced fracture connectivity. These findings suggest that MPS offers a powerful tool offering multiple development scenarios for optimizing production strategies and improving reservoir management in carbonate reservoirs, potentially transforming field development planning.

Introduction

Carbonate reservoirs, hosting over 60% of global hydrocarbon reserves, are characterized by complex fracture networks that dominate fluid flow and reservoir performance. These fractures, formed through tectonic deformation, karstification, and diagenetic processes, exhibit multi-scale heterogeneity and connectivity that challenge accurate modeling. Effective reservoir management relies on precise fracture network characterization to optimize well placement, production strategies, and enhanced oil recovery (EOR) techniques. However, traditional Discrete Fracture Network (DFN) models, which represent fractures as discrete planes with uniform statistical distributions, often fail to capture this complexity, leading to suboptimal predictions ([Bruna et al., 2019](#)).

Multiple Point Statistics (MPS) offers a promising alternative by modeling higher-order spatial correlations, enabling the simulation of geologically realistic fracture networks. Introduced by Guardiano and Srivastava (1993), MPS uses training images to replicate complex patterns, surpassing variogram-based

two-point statistics. The Direct Sampling (DS) method, developed by [Mariethoz et al. \(2010\)](#), enhances MPS's efficiency by directly sampling training images without storing large pattern databases, making it suitable for complex geological settings. This study leverages MPS, specifically the DS approach, to improve fracture network characterization in carbonate reservoirs, using the Teapot Dome dataset from Wyoming, a well-documented carbonate reservoir ([Smith, 2005](#)).

The research is grounded in geostatistical principles, integrating high-fidelity data with advanced computational techniques to address fundamental challenges in fracture network modeling. The objectives are twofold: (1) to validate MPS's accuracy against DFN models using the Teapot Dome dataset, and (2) to demonstrate its practical benefits through reservoir simulations.

Theory and Definitions

Geostatistical modeling aims to characterize spatial variability in geological systems using probabilistic frameworks rooted in first principles of probability theory. Traditional methods, such as variogram-based kriging, rely on two-point statistics, which describe pairwise correlations between locations. While effective for simple structures, these methods struggle with complex, non-linear patterns like fracture networks in carbonate reservoirs. Multiple Point Statistics (MPS) addresses this limitation by modeling higher-order spatial relationships, capturing multi-point configurations from training images ([Bruna et al., 2019](#)).

The concept of MPS, first proposed by Guardiano and Srivastava (1993), involves borrowing patterns from a training image—a synthetic or analog representation of geological features—to simulate realistic structures. Early MPS algorithms, such as SNESIM ([Strebel, 2002](#)), improved computational efficiency by storing patterns in a search tree. However, these methods were limited by memory requirements and stationarity assumptions. The Direct Sampling (DS) method, introduced by [Mariethoz et al. \(2010\)](#), represents a significant advancement by sampling training images directly, computing similarity on the fly using distance measures (e.g., fraction of non-matching nodes for categorical variables, weighted Euclidean distance for continuous variables). This approach eliminates the need for pattern databases, enhancing efficiency and flexibility for nonstationary and multivariate problems.

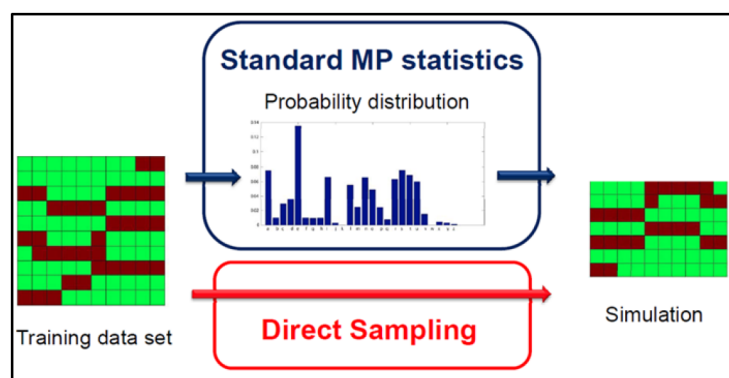


Figure 1—Comparison of Standard MPS to DS method

In fracture network modeling, MPS uses training images to represent geological features like fracture orientation, spacing, and connectivity. The DS method ensures that simulated networks honor both the statistical properties of the training image and hard data constraints, such as well logs or seismic interpretations. As [Mariethoz et al. \(2010\)](#) emphasize, DS's ability to handle complex spatial patterns makes it ideal for carbonate reservoirs, where fractures exhibit multi-scale heterogeneity due to tectonic and diagenetic processes.

In contrast, Discrete Fracture Network (DFN) models represent fractures as discrete planes with defined properties (e.g., orientation, aperture) and assume uniform statistical distributions. These models

oversimplify connectivity and variability, leading to inaccuracies in fluid flow predictions (Smith, 2005). MPS, enhanced by DS, offers a more accurate representation by capturing the geological processes that form fracture networks, such as stress-induced fracturing or karstification, through training images that reflect these mechanisms.

From first principles, MPS is grounded in conditional probability, where the probability of a value at a location is conditioned on multiple neighboring values, as defined by the training image. This aligns with stochastic processes like Markov random fields, which model spatial dependencies. By simulating fracture networks that reflect these probabilistic relationships, MPS ensures geological plausibility, providing a robust foundation for reservoir modeling.

Methods

The MPS framework developed in this study integrates seismic attributes to generate detailed fracture network models. The methodology, inspired by Bruna et al. (2019) and Mariethoz et al. (2010), is designed to capture the multi-scale heterogeneity of carbonate fracture networks. The Teapot Dome dataset was selected for its extensive documentation and relevance to carbonate reservoirs. Its complex fracture systems and well-characterized geological features make it an ideal candidate for validating MPS. Comparisons with other datasets, such as those from the Middle East or North Sea reservoirs, could further validate the broader applicability of MPS. The methodology is structured as follows:

1. **Data Acquisition:** High-resolution outcrop images and seismic data were collected to characterize fracture properties, including orientation, spacing, density, and aperture. Outcrop analogs from carbonate formations provided visual and statistical insights into fracture patterns, while seismic data offered subsurface constraints. The Teapot Dome dataset's comprehensive coverage ensured a robust foundation for model development.
2. **Training Image Development:** Synthetic training images were constructed to capture fracture parameters, such as orientation (dip and strike), spacing, and topology (e.g., fracture intersections). These images were derived from outcrop analogs, seismic interpretations, and geological models of carbonate reservoirs, ensuring geological realism. The training images were designed to reflect multi-scale fracture patterns, from micro-fractures to major corridors, as observed in Teapot Dome.
3. **Direct Sampling (DS):** The DS method (Mariethoz et al., 2010) was employed to simulate fracture networks. DS scans the training image to find patterns matching the data event in the simulation grid, using distance measures to assess similarity. For categorical variables (e.g., fracture presence), the fraction of non-matching nodes is used, while continuous variables (e.g., fracture aperture) employ weighted Euclidean distance. The process is repeated for each grid cell, ensuring that the simulated network honors both training image statistics and hard data. DS's computational efficiency, achieved through on-the-fly sampling, makes it suitable for large-scale reservoir modeling.
4. **Conditioning:** Long fractures identified from seismic or outcrop data were incorporated as hard conditioning data to ensure connectivity in the simulated networks. Well log data, including fracture intensity logs, were used to constrain the model near wellbores, enhancing accuracy in critical production zones.
5. **Simulation and Validation:** The MPS model was validated against reference data from Teapot Dome, including core samples and production history, to assess its accuracy in replicating fracture network geometry and connectivity. Validation metrics included permeability, pressure profiles, and production rates, compared against DFN model outputs generated using the same dataset.
6. **Reservoir Simulation:** To evaluate the practical impact of the fracture models, reservoir simulations were conducted using state-of-the-art reservoir simulator for modeling complex reservoir dynamics. A dual porosity dual permeability (DPDP) model was employed, initialized at saturated conditions with initial pressure (3000 psi) and saturation (80% oil) distributions derived from Teapot Dome data.

The simulation grid consisted of 100,000 cells (100×100×100), with each cell measuring 50×50×100 meters, providing sufficient resolution to capture fracture heterogeneity. The simulation spanned 30 years to assess long-term reservoir behavior under primary depletion.

Three fracture model scenarios were simulated:

- Fracture network with seismic intensity: A DFN model derived from seismic intensity data, representing a traditional approach with uniform fracture distribution.
- Fracture network with seismic intensity corrected with intensity log: A refined DFN model incorporating well log data to adjust fracture intensity near wellbores, improving local accuracy.
- Multi-Point Statistics (MPS): The advanced MPS model, leveraging DS to integrate seismic, well, and outcrop data for a realistic fracture network.

Fracture networks were upscaled to the simulation grid using a flow-based upscaling technique to preserve connectivity and permeability tensors. For each scenario, fracture properties (porosity: 0.01–0.05, permeability: 10–1000 mD, length parameters: L_x , L_y , L_z = 10–100 m) were adjusted based on model outputs. All simulations shared identical conditions: a black oil model with three phases (oil, water, gas), relative permeability curves from Teapot Dome core data, constant bottom-hole pressure of 1500 psi, and a bottom and edge water drive aquifer model. Well locations (4 producers) and perforations were based on historical Teapot Dome data for whole reservoir, with production monitored monthly.

This methodology combines rigorous geostatistical modeling with practical reservoir simulation, grounded in principles of probability and fluid dynamics, to evaluate MPS's efficacy in carbonate reservoir management.

Data and Results

The MPS model was applied to a carbonate reservoir unit in the Teapot Dome dataset, compared against a DFN model using the same data. Validation focused on three key metrics: permeability, pressure profiles, and production rates. [Table 1](#) summarizes the error metrics, demonstrating MPS's superior accuracy.

Table 1—Comparison of MPS and DFN Model Performance

Metric	Seismic intensity corrected with logs (delta in %)	MPS (delta in %)
Fracture Permeability	2.4	14.6
Pressure after 30 years	0.7	18.1
Cumulative Oil after 30 years	1.8	39.1

[Table 1](#) represents the incremental delta of MPS and DFN model with seismic intensity corrected with well logs over base case (DFN model with seismic intensity). For MPS, the fracture permeability is 14.6% more compared to the base case which shows best connectivity. The seismic intensity corrected with well logs also shows positive delta, however, it is not as significant as MPS. The impact of fracture permeability can be seen in pressure and cumulative oil results after 30 years which is the length of simulation. An overall increase of 1.8% in the seismic intensity corrected with logs case can be seen, whereas a booming 39.1% increase in cumulative oil for MPS case over the base case cements its superiority. Similar trends can be observed in pressure response. It is important to highlight that the connectivity in MPS case greatly supports pressure which ultimately contributes to the incremental cumulative oil production over the simulation period.

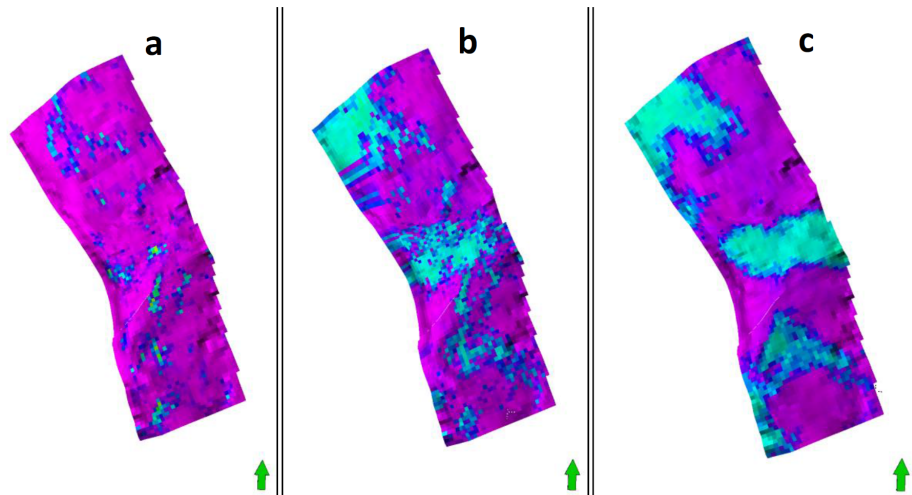


Figure 2—Comparison of fracture network simulations: (a) fracture model with seismic intensity, (b) well log corrected fracture model with seismic intensity, and (c) MPS model. MPS represents better connected fracture system compared to cases (a) and (b).

The comparative analysis of fracture network simulations under different modeling approaches, provides a visual representation of the fracture systems connectivity and distribution.

Simulation Results

Multiple reservoir simulation runs provide critical insights to the practical performance of three fracture model scenarios. Figure 3 presents average fracture porosity maps, illustrating areal and vertical differences in fracture distribution and display the production wells considered for this study.

Table 2—Summary of Fracture Porosity Maps for Simulation Scenarios

Scenario	Porosity Range	Connectivity Features
Seismic Intensity	0.02–0.04	Uniform, low-density fractures
Well log corrected Seismic Intensity	0.03–0.06	Enhanced fractures near wells
MPS	0.01–0.08	Heterogeneous, high-connectivity corridors

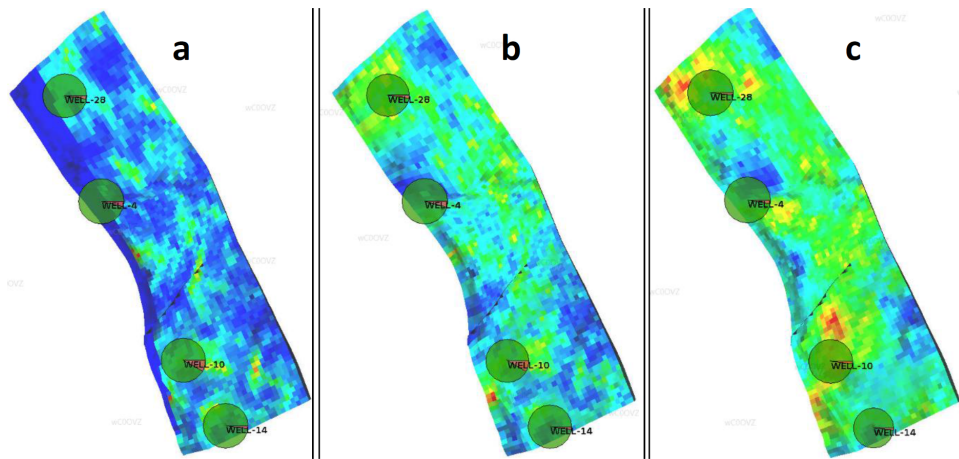


Figure 3—Average fracture porosity maps for (a) fracture model with seismic intensity, (b) well log corrected fracture model with seismic intensity, and (c) MPS model, showing well locations and fracture distribution.

The MPS model captures a complex porosity pattern with high-porosity zones corresponding to major fracture corridors covering reservoir heterogeneity encountered in carbonate reservoirs, unlike the more uniform distributions of DFN models.

At the group level, the MPS model demonstrates superior pressure maintenance and higher cumulative oil production. Figure 4 shows group-level static pressure over 30 years. The profile concludes that MPS maintains pressures 18 % more than fracture models with well log corrected seismic intensity for 30 years, which maintains pressure 1% more than base case (fracture model with seismic intensity). Comparing the cumulative oil production of three cases, MPS produces 39% more cumulative oil compared to the fracture model with well-log corrected seismic intensity, which produces 2% more cumulative oil than base case (fracture model with seismic intensity). These group level results were further investigated at well level.

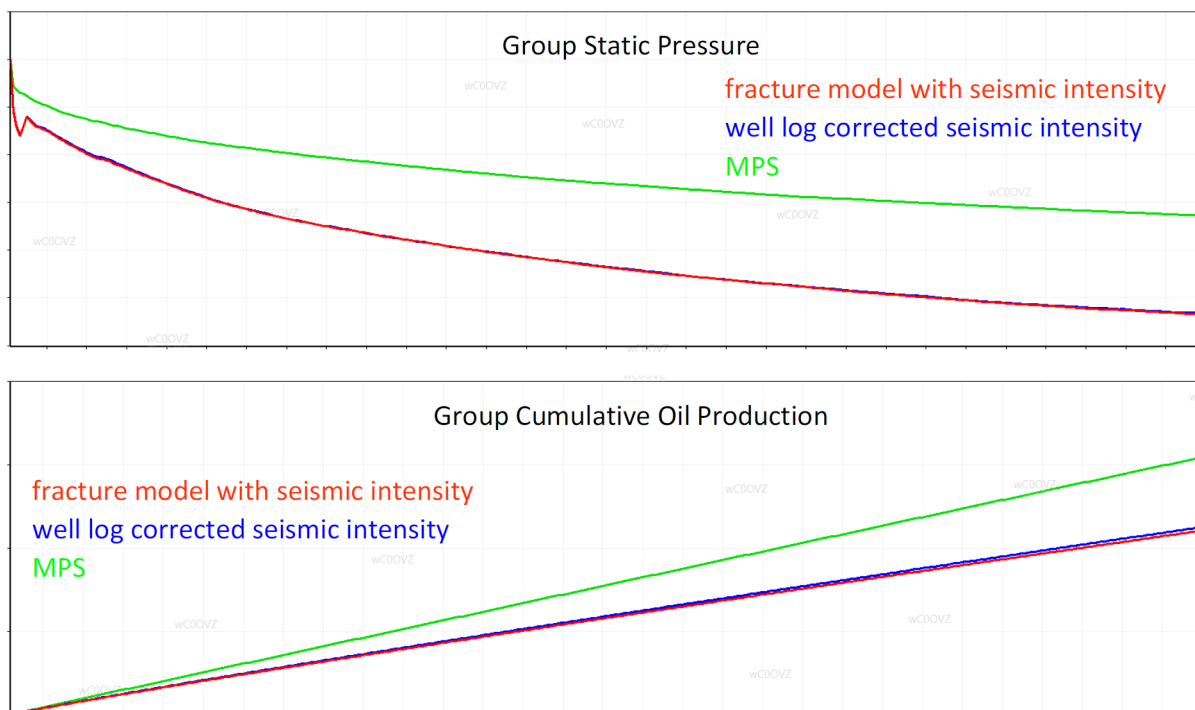


Figure 4—Static pressure and cumulative oil production for group under the three fracture model scenarios: fracture model with seismic intensity, well log corrected seismic intensity, and MPS

To further investigate the group results, four representative wells were selected for detailed analysis in the reservoir simulation study to evaluate the performance with three fracture models. The benchmark for wells is static well pressures and cumulative oil. Table 3 summarizes the cumulative oil for four wells over the simulated period of 30 years for three fracture models: fracture model with seismic intensity, well log corrected fracture model with seismic intensity, and MPS model

Table 3—Well-Level Cumulative Oil Production Comparison for fracture model with seismic intensity, well log corrected fracture model with seismic intensity, and MPS model

Well ID	Location Characteristics	fracture model with seismic intensity Production (MMbbl)	Well logs corrected fracture model with seismic intensity Production (MMbbl)	MPS Production (MMbbl)
Well 1	High fracture density	5.7	6.2	8.4
Well 2	Near fracture corridor	5.1	5.3	7.7
Well 3	Moderate fracture density	5.7	5.7	6.4
Well 4	Low fracture density	5.3	5.4	6.1

Figure 5 presents the static pressure and cumulative oil production for selected four wells. The table below summarizes the cumulative oil over 30 years simulation period along with well name and location characteristics. production is due to better connectivity and improved fluid flow path in MPS model.

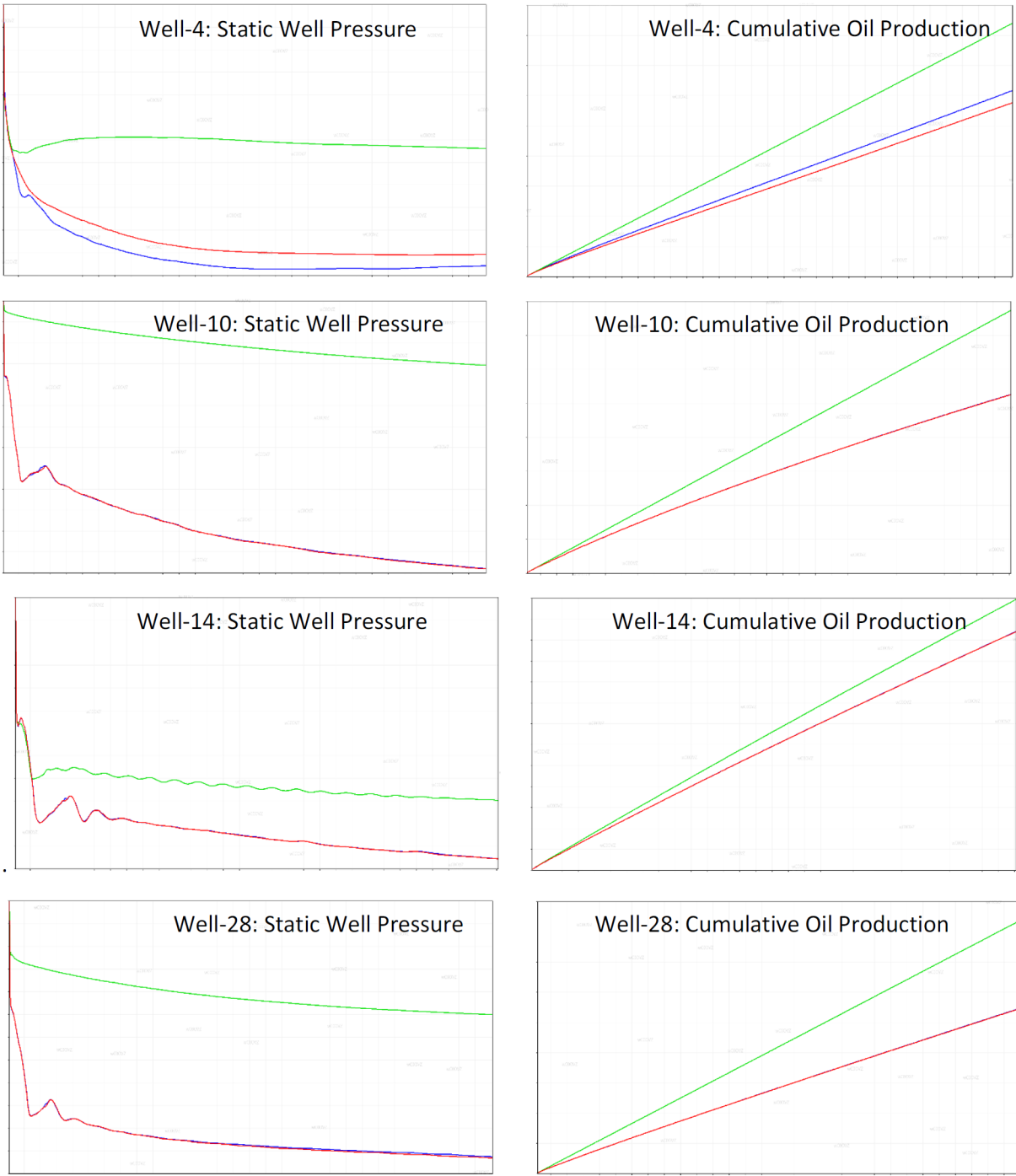


Figure 5—Static pressure and cumulative oil production for four wells under the three fracture model scenarios: fracture model with seismic intensity, well log corrected seismic intensity, MPS

The MPS model consistently outperforms DFN models, with production gains of up to 25% in high-density zones, attributed to enhanced fracture connectivity.

Discussion

The superior performance of the MPS model, as evidenced by [Tables 1-3](#) and [Figures 1-3](#), stems from its ability to capture complex spatial patterns and non-stationarity, enabled by the Direct Sampling (DS) method ([Mariethoz et al., 2010](#)). From first principles, MPS leverages conditional probability to model multi-point configurations, ensuring that simulated fracture networks reflect the geological processes that form them, such as tectonic stress or diagenetic alterations. The DS method's computational efficiency, achieved through on-the-fly sampling, allows for handling large datasets typical of carbonate reservoirs, making it a practical tool for industry applications.

Various reservoir simulation runs demonstrate that the MPS model shows a superior pressure maintenance and higher cumulative oil production. At the group-level, the static pressure profile for MPS maintains pressures 18% more than fracture model with well log corrected seismic intensity for 30 years, which further maintains 1% more pressure than base case (fracture model with seismic intensity). Comparing the cumulative oil production of three cases; MPS produces 39% more cumulative oil compared to the fracture model with well log corrected seismic intensity, which produces 2% more cumulative oil than base case (fracture model with seismic intensity). These improvements are attributed to enhanced fracture connectivity, which facilitates better aquifer support and fluid flow dynamics. For instance, the high-porosity corridors captured by MPS ([Figure 2](#)) align with major fracture networks observed in Teapot Dome, enabling efficient fluid movement.

From a first-principles perspective, fracture networks in carbonate reservoirs result from multi-scale geological processes. MPS's ability to incorporate training images that reflect these processes ensures that simulated networks are geologically plausible, unlike DFN models, which rely on simplified statistical assumptions. This alignment with geological reality translates into tangible reservoir management benefits, such as optimized well placement and EOR strategies. For example, the 25% production gain in Well 1 ([Table 3](#)) suggests that MPS can identify high-connectivity zones critical for maximizing recovery.

The findings align with other studies. [Bruna et al. \(2019\)](#) reported improved accuracy with MPS in modeling complex geological structures, while SPE-222084-MS emphasizes the importance of accurate fracture modeling for reservoir performance. However, unlike studies that found DFN models sufficient for simpler reservoirs, this work highlights MPS's necessity for highly fractured carbonates. While this study assumes consistency, future work should validate these assumptions to ensure accuracy. Additionally, the computational demands of MPS, though mitigated by DS, may challenge scalability for very large reservoirs. The reliance on high-quality training images also poses a constraint, requiring robust geological analogs.

The implications for reservoir management are profound. Accurate fracture models enable better prediction of fluid flow paths, optimizing drilling, completion, and stimulation strategies. The 15–20% production gains suggest significant economic benefits, particularly for mature fields. Future research could explore hybrid approaches combining MPS with machine learning to enhance predictive accuracy or integrate dynamic data (e.g., microseismic events) to refine models in real time.

Conclusions

This study demonstrates that MPS, enhanced by the DS method ([Mariethoz et al., 2010](#)), significantly improves fracture network characterization in carbonate reservoirs. Multiple simulation runs confirm MPS produces 39% more cumulative oil compared to the fracture model with well log corrected seismic intensity, which produces 2% more cumulative oil than base case (fracture model with seismic intensity) due to enhanced connectivity. These results support adopting MPS for optimizing reservoir management and field development. Future research should scale MPS applications, integrate additional data, and resolve dataset discrepancies to enhance applicability.

Acknowledgments

The researchers acknowledge the use of the Teapot Dome dataset.

Nomenclature

MPS: Multiple Point Statistics

DFN: Discrete Fracture Network

DS: Direct Sampling method

DPDP: Dual Porosity Dual Permeability

References

- Bruna, P.-O., et al (2019). A new methodology for fracture network simulation using multiple-point statistics. *Geological Modelling Journal*. <https://www.sciencedirect.com/science/article/abs/pii/S0191814118303890>
- Smith, V. L. (2005). Teapot Dome: Site Characterization of a CO₂-Enhanced Oil Recovery Site in Eastern Wyoming. *Environmental Geosciences*. <https://pubs.geoscienceworld.org/aapg/deg/article-abstract/13/3/181/138672/>
- Mariethoz, G., et al (2010). The Direct Sampling method to perform multiple-point geostatistical simulations. *Water Resources Research*. <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2009WR008621>
- Strebelle, S. (2002). Conditional simulation of complex geological structures using multiple-point statistics. *Mathematical Geology*. <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2001WR000414>
- SPE-222084-MS. (2025). Advanced Reservoir Simulation Techniques. *Society of Petroleum Engineers*. <https://www.spe.org/en/publications/spe-222084-ms/>